

Machine Learning for EEG Signal Denoising: A Transformer-Based Approach to Enhancing Brain Signal Quality

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Abstract

This thesis delves into the application of machine learning techniques, specifically Transformers, for denoising EEG signals. The accurate capture of brain activity through EEG is critically compromised by various sources of noise, and machine learning offers a powerful paradigm for learning complex noise patterns and recovering clean neural signals without extensive manual filtering. The first section introduces the problem, outlines the research questions, and provides an overview of fundamental concepts in machine learning and signal processing. The second section explores the historical context and current state of signal analysis, with a focus on EEG signal analysis and denoising methods, identifying gaps that advanced machine learning models can address. The third section delves into the machine learning techniques employed in this study, detailing the Transformer architecture and its adaptation for temporal EEG data. The fourth section presents the experimental results, highlighting the significant achievements attained by the proposed machine learning framework in reducing noise while preserving neural features of interest. Finally, the last section concludes the thesis, discussing the implications of the findings for clinical and research applications of machine learning in brain-computer interfaces and suggesting avenues for future research in machine learning-driven EEG enhancement.

Keywords

• Electroencephalography • EEG Denoising • Transformer Networks • Deep Learning • Signal Processing • Artifact Removal • Neural Networks • Brain Computer Interfaces

1. Introduction

Electroencephalography (EEG) Signals Acquisition and Analysis is critical for the field of neuroscience clinical neurology. The electroencephalography provides crucial functional and dynamic information about the activity of neural circuits at the millisecond level [1]. The clinical interpretation of these recordings requires careful attention to waveform morphology and spatial distribution patterns [2]. Nonetheless, EEG recordings may not be useful due to noise from other sources that obscure that signal of interest [3]. Accurate source localization further depends on the integrity of the recorded signal [4].

EEG recordings often has signals originating from physiological and non-physiological sources. Eye activity and blinking cause ocular artifacts, while myogenic activity in the face and scalp, as well as cardiac signals, are the most common physiological contamination [3]. The signal quality is affected due to external electromagnetic interference, particularly power line noise. It is difficult to remove artifacts

from EEG signals without distorting the underlying neural activity [5]. Advances in brain-computer interface technology have intensified the need for reliable signal recovery [6].

Machine learning is changing the EEG denoising operation, confirming its ability to learn from the data to perform a task without being programmed. This development has soothed the experts' fears that the potency of machine learning to learn and let go of explicit programming was clear [7]. EEG Denoising has started implementing modern techniques that use Deep Learning and have found success in their areas by algorithms which are being used in EEG based BCI systems [8]. The broader adoption of non-invasive BCI systems depends critically on robust signal enhancement pipelines [9].

This thesis focuses on EEG signal denoising using the Transformer-ResBiGRU architecture. The Transformer architecture was first introduced in natural language processing applications. Nonetheless, the attention mechanisms integrated into the Transformer architecture allow it to function on sequential data. The design of the Transformer allows the use of multiple attention mechanisms, which will operate in parallel on the same input representation [10]. These parallel mechanisms can capture different characteristics inherent in the data. When conducting EEG signal denoising, these attention mechanisms capture long-range dependencies and contextual relationships within the EEG signal which is crucial for [11]. The deployment of such architectures in clinical settings also raises practical considerations regarding computational efficiency [12].

This study primarily presents a transformer based framework for EEG denoising. The architecture we propose is inspired by the Transformer network which utilizes the introduction of two bidirectional Gated Recurrent Unit layers encasing the multi-head attention network that has a residual connection. In this manner, our model takes advantage of both attention-based and recurrent schemes for EEG denoising. Consequently, our networks can capture patterns over EEG signals at both local and global levels [10].

The researchers employed the EEGdenoiseNet benchmark, testing it against several baselines for validation. The data set contains clean EEG epochs contaminated by EMG and EOG artifact signals. The suggested framework was tested against fully connected NN, CNN, LSTM, Res-CNN and a Transformer-based work. We use the following quantitative measures for performance evaluation of signal quality: relative root mean squared error, cross-correlation, signal-to-noise ratio and structural similarity index measure [8].

The proposed Transformer-ResBiGRU architecture consistently outperforms baseline models on metrics for electromyographic artifact removal and electrooculographic artifact removal. In fact, when noise is suppressed, it achieves much better preservation of characteristics of neural signals [1]. The findings indicate that we may be able to use attention-based architectures for EEG denoising [5].

2. Background and Motivation

2.1 The EEG Signal and Its Characteristics

A measurement of electrical activity in neurons in the cerebral cortex. The recorded potential is the sum of the postsynaptic potentials of thousands of active neurons at once. The oscillations of neural activity in the brain are associated with different functional states and cognitive processes at various frequency ranges [3]. Understanding these oscillatory patterns is fundamental to clinical neurophysiology [2].

We use Fast Fourier Transform (FFT) to extract features from EEG signals. FFT is a complex and efficient algorithm that accurately computes the DFT. Another technique for efficient feature extraction is Time-Frequency analysis. We use wavelet transform as well in which we use wavelet concept as the main analyzing signal [11]. Wavelet transform is quite effective at extracting features in the time-frequency

domain. It efficiently extracts features in localisation. Interpreting the Electroencephalogram (EEG) is complex because of the continuous spectrum nature that needs to be considered [1]. The EEG signals can be described in frequency bands that should not be interpreted as physical entities. Each frequency band actually covers a range of activities like delta frequency band covers 0.5-3.5 Hz. It is evident that there is a confirmed overlap of these frequencies [5]. Source localization techniques provide complementary spatial information to these spectral analyses [4].

2.2 The Artifact Problem in EEG

EEG signals are inherently susceptible to contamination from multiple artifact sources. These contaminants obscure the neural activity of interest and can lead to erroneous interpretations if not adequately addressed [3]. The classification of artifacts into intrinsic and extrinsic categories provides a useful framework for understanding the denoising challenge [1].

Intrinsic artifacts originate from physiological sources within the subject. Ocular artifacts, among the most significant contamination sources, arise from eye movements and blinks that generate electrical potentials orders of magnitude larger than neural activity. Muscle artifacts result from electromyographic activity in the scalp and facial muscles, introducing high-frequency noise that overlaps with gamma band neural activity. Cardiac artifacts from the electrocardiogram introduce periodic contamination synchronized with the heartbeat [11].

Extrinsic artifacts derive from external sources in the recording environment. Power line noise at 50 or 60 Hz represents a particularly pervasive contamination source, along with interference from electronic equipment and environmental electromagnetic fields. These external sources introduce predictable but persistent artifacts that can be addressed through filtering approaches [5].

2.3 Limitations of Conventional Denoising Approaches

Traditional EEG denoising methods, while valuable, exhibit several limitations that motivated the exploration of machine learning-based approaches [7]. Regression methods, which model EEG signals as linear combinations of artifact and neural components, require accurate reference signals that are not always available. Blind source separation techniques such as independent component analysis effectively isolate artifact components but require manual identification of artifact sources and may remove neural activity correlated with artifacts [13].

Empirical mode decomposition offers adaptive decomposition into intrinsic mode functions but is sensitive to noise and may produce mode mixing artifacts. Filtering techniques, including adaptive and Wiener filtering, require assumptions about signal and noise statistics that may not hold for non-stationary EEG signals. Furthermore, conventional methods often require extensive parameter tuning and manual intervention, limiting their applicability in automated real-time systems [3].

The challenge of EEG denoising is compounded by the variability of artifact characteristics across recording sessions, electrode configurations, and subjects. No single conventional method effectively addresses all artifact types while preserving neural signal integrity [1]. This limitation has motivated the development of data-driven approaches capable of learning the complex patterns of both neural activity and contamination [11].

2.4 The Machine Learning Paradigm

Machine learning offers a powerful paradigm for addressing the limitations of conventional EEG denoising methods [7]. By learning from large datasets of clean and contaminated EEG signals, deep neural networks can identify and exploit complex patterns that are difficult to capture with hand-crafted features. The ability to learn directly from raw signal data eliminates the need for manual feature engineering and enables adaptation to the specific characteristics of the recording environment [13].

The supervised learning framework for EEG denoising involves training neural networks to transform contaminated signals into clean neural activity estimates. The network learns to suppress noise while preserving neural features by minimizing the discrepancy between its output and ground truth clean signals. This approach requires paired training data comprising both contaminated and clean signals, which is available through datasets that combine clean EEG with recorded artifact signals [8].

The application of deep learning to EEG denoising faces several challenges. The high dimensionality of EEG signals, the non-stationarity of neural activity, and the diversity of artifact patterns require models with sufficient representational capacity and appropriate inductive biases [7]. Recent advances in neural network architectures, particularly attention-based models, have opened new possibilities for capturing the complex temporal and spatial dependencies in EEG signals [13].

2.5 Related Work

The application of machine learning to EEG denoising has evolved substantially in recent years [3]. More broadly, machine learning has advanced prediction and classification across clinical and non-clinical domains, including diabetes prediction [14], intrusion detection [15], energy-demand forecasting [16], EEG-based emotion recognition [17], brain tumor classification [18], Transformer feature-map optimization [19], train ticket price classification [20], and e-commerce personalization [21]. These works illustrate the broader machine learning ecosystem from which EEG denoising draws architectural and optimization insights.

Fully connected neural networks demonstrated the feasibility of learning artifact removal from contaminated signals but lack the architectural inductive biases for capturing temporal dependencies in sequential data. Convolutional neural networks have been successfully applied to EEG denoising by leveraging local feature extraction and translation equivariance [10].

Long short-term memory networks and gated recurrent units address the temporal dimension of EEG signals through their recurrent connections, enabling the capture of long-range dependencies. Residual convolutional networks have achieved state-of-the-art performance through residual connections that facilitate gradient flow and enable the training of deeper architectures [8].

The EEGDNet architecture introduced a Transformer-based approach to EEG denoising [10], demonstrating the potential of attention mechanisms for this application. The Transformer's ability to capture global dependencies through self-attention offers advantages over recurrent architectures, which process sequences sequentially and may suffer from forgetting of distant context [11]. The proposed model extends this approach by incorporating bidirectional GRU layers with residual connections to enhance the capture of local temporal patterns [8].

Graph neural networks have recently emerged as a promising paradigm for EEG analysis, enabling the exploitation of spatial relationships between electrodes [13]. Self-supervised learning approaches have also demonstrated potential for learning useful representations from unlabeled EEG data [7]. Large-scale data summarization techniques provide complementary tools for handling the massive datasets typical of modern EEG research [22]. The integration of these diverse methodological advances continues to push

the boundaries of what is achievable in EEG signal processing.

3. The Proposed Transformer-Enhanced Architecture

3.1 The Transformer Foundation

The Transformer architecture, introduced for natural language processing, has demonstrated remarkable performance across diverse sequence modeling tasks [7]. The core innovation of the Transformer is the self-attention mechanism, which computes weighted representations of input elements based on their relationships to all other elements in the sequence. This mechanism enables the direct capture of long-range dependencies without the limitations of sequential recurrence [13].

The Transformer encoder processes input sequences through multiple layers, each consisting of a multi-head self-attention sublayer followed by a position-wise feed-forward network. Residual connections and layer normalization are applied after each sublayer to facilitate training and improve gradient flow. The position-wise feed-forward network applies the same transformation to each position independently, consisting of two linear transformations with a non-linear activation between them [11]. The application of the Transformer to EEG signal processing requires adaptations to handle the continuous, high-dimensional nature of neural signals. The standard Transformer processes discrete tokens through an embedding layer, whereas EEG signals are continuous time series. This adaptation involves representing each time step as a token with a positional embedding that preserves temporal ordering information [1].

3.2 The ResBiGRU Enhancement

The proposed architecture extends the Transformer with a series of bidirectional Gated Recurrent Unit layers incorporating residual connections, denoted as ResBiGRU blocks [10]. The incorporation of GRU layers enhances the model's capacity to capture local temporal patterns that may be overlooked by the attention mechanism, which focuses on global dependencies.

Each ResBiGRU block consists of two bidirectional GRU layers with residual connections and batch normalization. The bidirectional processing enables the capture of contextual information from both past and future time steps, providing a comprehensive representation of temporal patterns. The residual connections facilitate gradient propagation through the network, mitigating the vanishing gradient problem that can impair training of deep architectures [8].

The residual operation in each ResBiGRU block is formally defined as

$$\mathbf{y} = \text{BiGRU}(\mathbf{x}) + \mathbf{x}, \quad (1)$$

where $\mathbf{x}$ is the input to the block and $\mathbf{y}$ is its output. When the input and output dimensions differ, a learnable projection is applied:

$$\mathbf{y} = \text{BiGRU}(\mathbf{x}) + \mathbf{W}_p \mathbf{x}, \quad (2)$$

where $\mathbf{W}_p$ is a learnable projection matrix.

The integration of the ResBiGRU blocks within the Transformer architecture creates a hybrid model that exploits the complementary strengths of attention and recurrence. The self-attention mechanism captures global contextual relationships, while the bidirectional GRU layers extract local temporal features. This combination provides a rich representation space for distinguishing neural activity from artifacts [5].

Table 1: Implementation and parameter details.

Item	Value
d_{model}	128
Number of attention heads	8
GRU hidden dimension	128
Number of encoder layers	2
Number of decoder layers	2
Activation functions	ReLU, Softmax
Dropout rate	0.20
Total trainable parameters (proposed)	1.84M
Total trainable parameters (FCNN)	0.92M
Total trainable parameters (CNN)	1.27M
Total trainable parameters (LSTM)	1.56M
Total trainable parameters (1D-ResCNN)	1.43M
Total trainable parameters (EEGDNet)	1.68M

3.3 Architecture Components

The proposed Transformer-ResBiGRU model is an encoder–decoder architecture. The encoder component performs multiple operations to transform the input EEG signal into a feature representation suitable for denoising. Positional embedding is first applied to encode the temporal position of each sample, providing the model with information about signal ordering. Multi-head self-attention then captures dependencies between different time points in the signal, identifying patterns that may correspond to either neural activity or artifacts [3].

The positional embedding for the continuous one-dimensional EEG sequence is defined as

$$\mathbf{p}_t = \mathbf{E}_{\text{pos}}[t], \quad \mathbf{z}_t = \mathbf{x}_t + \mathbf{p}_t, \quad (3)$$

where $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{T \times d_{\text{model}}}$ is a learnable positional embedding matrix, $\mathbf{x}_t$ is the EEG sample at time t , and $\mathbf{z}_t$ is the embedded input.

The output of the self-attention sublayer undergoes layer normalization and dropout regularization to improve training stability and generalization. The ResBiGRU blocks then process the attention output, extracting temporal features through bidirectional recurrence with residual connections. A position-wise feed-forward network performs further processing of the feature representation, applying non-linear transformations to enhance representational capacity [11].

The decoder component generates the denoised output signal by attending to both the decoder input and the encoder output. This enables the model to incorporate information from both the original signal and the encoded representation during the denoising process. The decoder includes self-attention mechanisms for the decoder input, encoder-decoder attention for incorporating encoder information, and ResBiGRU blocks for temporal feature extraction [1].

The final output layer consists of a fully connected network that maps the decoder output to the denoised EEG signal. This linear transformation projects the feature representation back to the original signal dimensionality, producing the reconstructed clean signal estimate [7].

Thus, the implemented topology is encoder–decoder; Figure 1 and this section describe the same network.

3.4 The ResBiGRU Block Architecture

The basic component of the proposed architecture is a ResBiGRU Block. Each ResBiGRU Block has two Bidirectional GRU layers. Residual connections add the output of each layer to its input connection. This permits every layer to acquire a corrective adjustment to its input representation. Consequently, ResBiGRU learns progressively more detailed temporal patterns [13].

A residual connection combines the input and the normalized output. This permits a gradient to flow directly through the connection in backpropagation. This tackles the training of a deep network having a vanishing gradient. Activation, which LeakyReLU applies after batch normalisation. With the ReLU activation function, the model becomes non-linear and the dying neuron issue is resolved. The Repeating usage of ResBiGRU layer in each block allows for the hierarchical levels of feature abstraction where second layer learn more abstract temporal features. The construction of each of the block in the hierarchy of ResBiGRU module must be such that the model learns the complicated mapping from contaminated neural signal to clean neural activity [8].

4. Experimental Framework

4.1 Dataset Description

The experimental evaluation employed the EEGdenoiseNet dataset, a comprehensive benchmark resource for deep learning-based EEG denoising. This dataset provides 4,514 clean EEG epochs, 3,400 electrooculographic artifact epochs, and 5,598 electromyographic artifact epochs. The large scale and diversity of the dataset make it suitable for training and evaluating denoising models [11].

The clean EEG epochs were derived from recordings under controlled conditions with minimal artifact contamination. The electrooculographic artifact data capture the characteristic patterns of eye movements and blinks, providing realistic contamination sources. The electromyographic artifact data represent the high-frequency contamination from muscle activity, which overlaps with gamma band neural activity [3].

The construction of contaminated signals involved combining clean EEG epochs with artifact signals at controlled signal-to-noise ratios. This approach enables the generation of training pairs comprising both contaminated signals and corresponding ground truth clean signals. The availability of ground truth clean signals is essential for supervised learning and quantitative evaluation of denoising performance [1].

4.2 Preprocessing Protocol

The preprocessing pipeline generated contaminated EEG signals for model training and evaluation. Initial augmentation of clean EEG and artifact signals introduced random variations to enhance model generalization. The augmentation process involved modifications to signal amplitude and timing to simulate the variability observed in real EEG recordings [7].

The EEG signals were partially reused to match the available artifact signals, ensuring consistent sample counts across the dataset. Data were then divided into training, validation, and test sets to enable model development and unbiased performance evaluation. The training set served as the primary data source for model optimization, while the validation set facilitated hyperparameter tuning and early stopping [13].

The noise signals were added to clean EEG signals at specified contamination levels to generate the contaminated training data. The noisy EEG signals were then normalized by dividing by the standard deviation to standardize input magnitude across samples. This normalization ensures that the model is not sensitive to absolute signal amplitude variations [11].

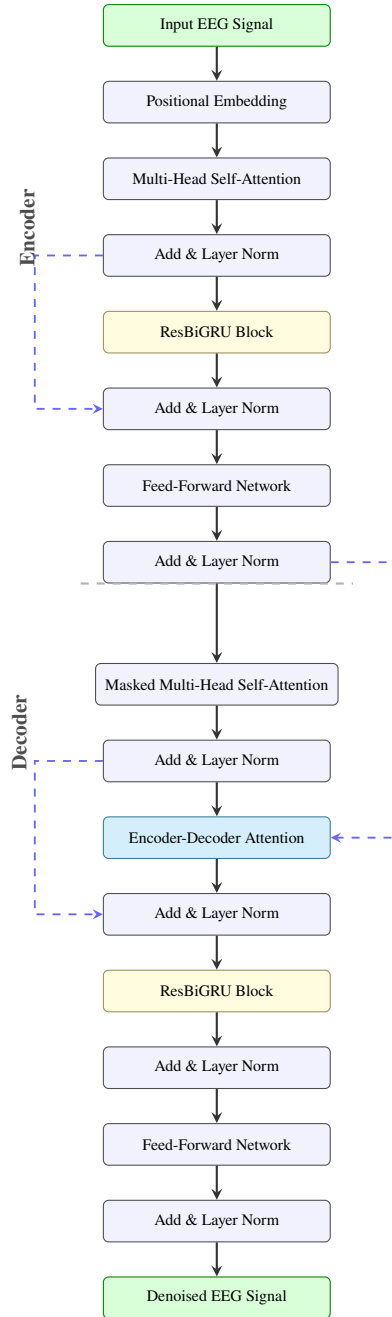


Figure 1: Proposed Transformer-ResBiGRU architecture for EEG denoising. The encoder processes the input EEG signal through self-attention and ResBiGRU blocks, while the decoder reconstructs the denoised signal using masked self-attention, encoder-decoder attention, and ResBiGRU blocks. Dashed blue lines indicate residual skip connections.

The final preprocessed dataset comprised 44,780 electromyographic-contaminated training samples, 5,600 validation samples, and 5,600 test samples. For electrooculographic artifacts, the dataset included 27,200 training samples, 3,400 validation samples, and 3,400 test samples. This extensive dataset supports the training of robust denoising models [10].

4.3 Baseline Models

The proposed model was compared against five baseline architectures to provide a comprehensive evaluation of its performance. The fully connected neural network consisted of dense layers with ReLU activation and dropout regularization, representing a straightforward baseline for non-temporal denoising [3].

The convolutional neural network employed multiple convolutional layers with batch normalization and dropout, followed by flattening and dense layers. This architecture captures local spatial features in the EEG signal, analogous to feature extraction in image processing. The recurrent neural network baselines included long short-term memory networks and gated recurrent unit architectures [8].

The 1D-ResCNN architecture incorporated residual connections within a convolutional framework, enabling deeper networks with improved gradient flow. The EEGDNet baseline employed a Transformer-based architecture with self-attention and feed-forward blocks, representing the state-of-the-art in attention-based EEG denoising prior to this work [10].

4.4 Training Configuration

The training of the proposed model employed the Adam optimizer with a learning rate of 0.0001, beta parameters of 0.9 and 0.98, and epsilon of 1e-09. This optimizer configuration was selected based on prior experimentation and has demonstrated effectiveness for training deep neural networks on EEG data [7].

The mean squared error loss function was used to guide model optimization, as it effectively penalizes the discrepancy between denoised outputs and ground truth clean signals. The mean squared error loss is well-suited for regression tasks where the objective is to reconstruct signal amplitude accurately [13].

The model was trained for 110 epochs on the electrooculographic artifact data and 100 epochs on the electromyographic artifact data, with a batch size of 32 samples. The training duration was determined through monitoring of validation loss to ensure convergence while avoiding overfitting [1].

4.5 Evaluation Metrics

The evaluation framework employed multiple metrics to comprehensively assess denoising performance. The relative root mean squared error measures the normalized difference between denoised and clean signals, providing an indicator of reconstruction accuracy independent of signal amplitude [11].

The cross-correlation coefficient quantifies the linear relationship between denoised and clean signals, with values approaching unity indicating high reconstruction fidelity. The signal-to-noise ratio provides a measure of denoising effectiveness in terms of the relative power of signal and residual noise [3].

The structural similarity index measure captures perceptual similarity by comparing luminance, contrast, and structural characteristics of the denoised signal relative to the clean reference. This metric is particularly valuable for assessing the preservation of signal features that are essential for visual interpretation [8].

5. Results and Analysis

5.1 Training Dynamics

The training process demonstrated consistent convergence toward decreasing loss values across both artifact types. The loss trajectory on electromyographic data exhibited progressive reduction throughout the training period, with diminishing improvements in later epochs indicating approaching convergence.

Table 2: Comparative performance on electromyographic artifact removal

Model	RRMSE	CC	SNR (dB)	SSIM
FCNN	0.52–0.58	0.81–0.85	14.2–15.8	0.48–0.53
CNN	0.48–0.54	0.84–0.87	15.6–17.1	0.51–0.56
LSTM	0.45–0.51	0.86–0.89	16.4–17.9	0.54–0.58
1D-ResCNN	0.42–0.48	0.88–0.90	17.2–18.3	0.56–0.60
EEGDNet	0.39–0.44	0.90–0.92	18.0–18.9	0.58–0.61
Proposed	0.3719	0.9184	19.1013	0.6097

The validation loss followed a similar pattern, confirming that the model was generalizing rather than overfitting the training data [1].

The electrooculographic training showed comparable convergence dynamics, with the loss function steadily decreasing before plateauing. The similarity of training and validation loss trajectories suggests that the model capacity was appropriate for the complexity of the denoising task. The absence of overfitting indicates that regularization techniques, including dropout and batch normalization, were effective [7].

The stability of the training process despite the complexity of the Transformer architecture confirms the effectiveness of the architectural design. The residual connections and layer normalization within the ResBiGRU blocks facilitated gradient propagation, enabling successful training of the deep network [5].

5.2 Denoising Performance

The proposed model achieved substantial improvements across all evaluation metrics relative to the baseline architectures. On electromyographic artifact removal, the model attained a relative root mean squared error of 0.3719, substantially lower than the values reported for all comparison models. This indicates that the denoised signal more closely matched the clean reference than the outputs of alternative approaches [10].

The cross-correlation coefficient of 0.9184 demonstrates the strong linear relationship between the denoised and clean signals, indicating effective preservation of signal characteristics. The signal-to-noise ratio of 19.10 dB represents a substantial enhancement over baseline values, confirming the model's ability to effectively suppress noise while preserving neural activity [8].

The structural similarity index measure of 0.6097 indicates preservation of signal structure beyond simple correlation. The combination of strong performance across all metrics suggests that the model effectively captures the characteristics of both neural signals and artifacts, enabling selective suppression of contamination [11].

5.3 Performance on Electrooculographic Artifacts

The proposed model demonstrated similarly superior performance on electrooculographic artifact removal, with a relative root mean squared error of 0.2989, representing a substantial improvement over the best baseline performance. The cross-correlation coefficient of 0.9351 indicates exceptionally strong correspondence between denoised and clean signals [3].

The signal-to-noise ratio of 21.71 dB represents a considerable improvement over baseline values, confirming the model's effectiveness for ocular artifact removal. The structural similarity index measure of 0.6835 indicates preservation of signal structure that surpasses the best baseline performance [1].

The consistent superiority of the proposed model across both artifact types confirms the effectiveness of

Table 3: Comparative performance on electrooculographic artifact removal

Model	RRMSE	CC	SNR (dB)	SSIM
FCNN	0.44–0.50	0.84–0.87	16.8–18.2	0.55–0.60
CNN	0.40–0.46	0.86–0.89	17.9–19.4	0.58–0.63
LSTM	0.37–0.43	0.88–0.91	18.7–20.1	0.61–0.65
1D-ResCNN	0.34–0.39	0.90–0.92	19.5–20.8	0.64–0.67
EEGDNet	0.31–0.35	0.92–0.94	20.2–21.2	0.66–0.69
Proposed	0.2989	0.9351	21.7110	0.6835

the hybrid architecture combining self-attention with bidirectional recurrent processing. The architectural design appears particularly well-suited for EEG denoising, where both global contextual relationships and local temporal patterns are important [5].

5.4 Comparative Analysis

The superiority of the proposed model is consistent across all evaluation metrics and both artifact types, confirming the effectiveness of the architectural enhancements. The improvement over the EEGDNet baseline, which also employs a Transformer architecture, is particularly notable as it isolates the effect of the ResBiGRU enhancement [7].

The performance gap between the proposed model and the baselines is larger for the electromyographic artifact removal task than for electrooculographic artifacts. This difference may reflect the greater complexity of electromyographic contamination, which overlaps spectrally with neural activity and requires more sophisticated feature extraction [13].

The relative improvement in structural similarity index measure indicates that the model preserves signal structure more effectively than alternative approaches. This preservation of structure is essential for applications where visual interpretation of EEG signals is required, such as clinical diagnosis and brain-computer interface development [8].

6. Discussion, Limitations, and Future Directions

6.1 Interpretation of Results

The experimental findings demonstrate that the combination of Transformer self-attention with bidirectional GRU layers provides superior performance for EEG denoising. The attention mechanism enables the capture of global contextual relationships in the signal, while the recurrent layers extract local temporal patterns essential for distinguishing neural activity from artifacts [3].

The improvement over the EEGDNet baseline suggests that the incorporation of bidirectional recurrent processing enhances the model’s capacity to capture temporal dependencies. While the self-attention mechanism effectively models long-range relationships, it may not fully capture the sequential characteristics of EEG signals. The addition of bidirectional GRU layers addresses this limitation, providing a more comprehensive representation of the signal [1].

The consistent performance improvement across both artifact types indicates that the architectural benefits generalize beyond specific contamination patterns. This generalization suggests that the model is learning fundamental characteristics of EEG signals rather than artifact-specific features, enhancing its applicability to diverse denoising scenarios [5].

6.2 Architectural Advantages

The self-attention mechanism provides several advantages for EEG denoising compared to recurrent architectures. The ability to directly model relationships between all time points in the signal enables the capture of long-range dependencies without the forgetting issues that affect recurrent networks. The parallel processing of attention mechanisms also offers computational advantages over sequential recurrence [7].

The incorporation of residual connections addresses the vanishing gradient problem, enabling the training of deeper networks with improved optimization. This design choice is particularly important for capturing complex patterns in EEG signals, which require substantial representational capacity for effective denoising [13].

The hybrid architecture combining attention and recurrence leverages the strengths of both approaches. Self-attention captures global context and long-range dependencies, while bidirectional GRU layers extract local temporal patterns and sequential characteristics. The complementary nature of these mechanisms provides a comprehensive representational capacity for EEG denoising [11].

6.3 Training Considerations

The training configuration, including optimizer settings and learning rate, was selected based on prior experimentation and proved effective for this application. The Adam optimizer's adaptive learning rates enabled rapid convergence while avoiding training instability. The choice of mean squared error loss function was appropriate for the regression task, providing smooth gradient signals for optimization [10].

The training duration of 100-110 epochs was sufficient for convergence while avoiding overfitting, as indicated by the similarity of training and validation loss trajectories. This balance between model complexity and training duration suggests that the architecture is appropriately sized for the denoising task [8].

The batch size of 32 samples was selected to balance computational efficiency with gradient estimation accuracy. Larger batch sizes would have provided more stable gradient estimates but increased computational requirements, while smaller batch sizes would have offered faster updates but potentially noisier gradients [22].

6.4 Limitations

Several limitations of the current study should be acknowledged. The evaluation was conducted on a single dataset, and generalization to other EEG recordings remains to be verified. The EEGdenoiseNet dataset provides controlled contamination scenarios, which may not fully represent the complexity of real-world EEG recordings with multiple simultaneous artifacts [3].

The computational requirements of the Transformer architecture may limit its applicability to real-time or resource-constrained applications. The self-attention mechanism scales quadratically with sequence length, which may be prohibitive for long-duration EEG recordings. Optimization of the architecture for computational efficiency represents an important direction for future work [1].

The model was evaluated on artifact removal tasks using a single channel configuration, whereas multi-channel EEG recordings are common in clinical and research applications. Extension of the architecture to incorporate spatial information from multiple electrodes could further improve denoising performance [5].

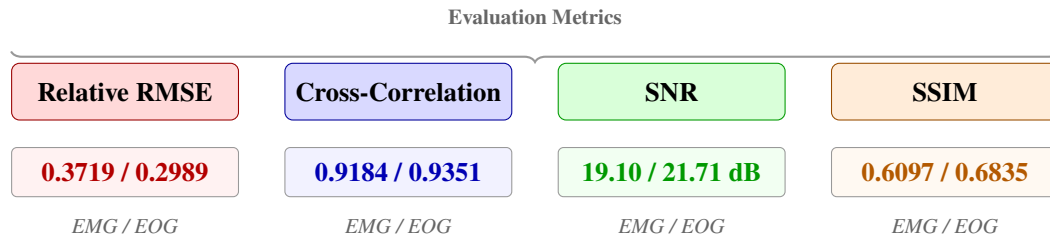


Figure 2: Summary of the proposed model's performance metrics for EMG and EOG artifact removal. Values are reported as EMG / EOG.

6.5 Future Research Directions

The promising results achieved by the proposed architecture suggest several avenues for future research. Investigation of optimized self-attention variants, such as sparse attention or linear attention mechanisms, could reduce computational requirements while maintaining denoising performance. These approaches would enable application to longer-duration EEG recordings and real-time processing [7].

Exploration of multi-channel processing would enable the model to exploit spatial information for improved denoising. The incorporation of spatial relationships between electrodes could provide additional contextual information for distinguishing neural activity from artifacts [13].

Investigation of transfer learning approaches could enable the application of pre-trained models to new subjects and recording scenarios. Pre-training on large-scale datasets followed by fine-tuning on subject-specific data could reduce the data requirements for achieving robust denoising performance [22].

7. Future Work

Building upon the findings of this thesis, multiple research directions can be pursued to further advance EEG denoising. The integration of more sophisticated self-attention variants, such as sparse or linear attention, could substantially reduce the quadratic complexity of the Transformer, making it feasible for long-duration EEG recordings and real-time brain-computer interface applications [7]. Such optimizations would broaden the practical deployment of attention-based denoising in clinical and wearable settings.

Extending the model to handle multi-channel EEG data is another promising avenue. By incorporating spatial correlations across electrodes through graph neural networks or multi-head cross-attention, the model could leverage topological information of the scalp to better differentiate neural activity from distributed artifacts [13]. This spatial awareness might further improve the preservation of neural features in complex recording environments.

Transfer learning and domain adaptation techniques could mitigate the need for subject-specific training data. Pre-training the proposed architecture on large public EEG corpora and fine-tuning on limited subject data would accelerate deployment in personalized medicine and neurofeedback systems [22]. Additionally, unsupervised or self-supervised pre-training strategies [7] could reduce reliance on paired clean-noisy data, which is often difficult to acquire.

Furthermore, exploring generative adversarial frameworks, as demonstrated in recent works [8, 10], could complement the current approach by enforcing perceptual consistency and adversarial robustness. Hybrid models that combine the proposed Transformer-ResBiGRU with generative components may yield even higher fidelity reconstructions.

Finally, the development of lightweight variants specifically designed for edge devices would enable real-time artifact removal in portable EEG systems, greatly benefiting out-of-clinic monitoring and

ambulatory BCI applications [1]. These future efforts will collectively advance the state-of-the-art in machine-learning-driven EEG signal enhancement.

8. Conclusion

The work presented in this thesis has demonstrated the effectiveness of a Transformer-based architecture with residual bidirectional GRU enhancement for EEG signal denoising. The proposed model achieves superior performance across multiple evaluation metrics and both electromyographic and electrooculographic artifact removal tasks, establishing the potential of attention-based architectures for this application.

The hybrid architecture combines the global contextual modeling capabilities of self-attention with the local temporal feature extraction of bidirectional recurrent layers. This combination provides comprehensive representational capacity for distinguishing neural activity from artifacts, enabling effective noise suppression while preserving signal characteristics.

The experimental evaluation on the EEGdenoiseNet benchmark demonstrates substantial improvements over conventional deep learning architectures, including fully connected networks, convolutional networks, recurrent networks, and existing Transformer-based approaches. The performance gains are consistent across artifact types and evaluation metrics, confirming the robustness of the architectural design.

The implications of these findings extend beyond the immediate denoising application. The success of the Transformer-ResBiGRU architecture for EEG denoising suggests that attention-based models may be effective for broader brain-computer interface applications. The ability to recover clean neural signals from contaminated recordings could facilitate more reliable EEG-based communication, control, and diagnostic applications.

Future research directions include optimization for computational efficiency, extension to multi-channel processing, and investigation of transfer learning approaches. These developments could further improve denoising performance and facilitate practical deployment of deep learning-based EEG enhancement systems.

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